THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The 10th term of $21, 18, 15, 12, \ldots$ is: 1

(a) -9 **(b)** -30 **(c)** -3 **(d)** -6

2. In $\triangle ABC$, $DE \parallel BC$, $AD = 1$ cm, $DB = 3$ cm, $AE = 2$ cm. Then EC (in cm) $=$ 1

(a) 2 **(b)** 7 **(c)** 3 **(d)** 6

3. The distance between $(2, 3)$ and $(5, 7)$ is: 1

(a) 1 **(b)** 7 **(c)** 25 **(d)** 5

4. The 5th term from the end of the AP with last term 100 and $d = 4$ is: 1

(a) 116 **(b)** 84 **(c)** 88 **(d)** 80

5. $\triangle ABC \sim \triangle DEF$ with areas 16 cm² and 36 cm². If $AB = 6$ cm, then DE (cm) $=$ 1

(a) 12 **(b)** 6 **(c)** 9 **(d)** 27

6. The point which divides the join of $(0, 0)$ and $(6, 9)$ in the ratio 1:2 internally is: 1

(a) $(2, 3)$ **(b)** $(4, 6)$ **(c)** $(6, 9)$ **(d)** $(3, \frac{9}{2})$

7. If $a_n = 3n + 2$, then the common difference is: 1

(a) 6 **(b)** 2 **(c)** 3 **(d)** 5

8. The length of the diagonal of a square of side 3 cm, squared, is: 1

(a) 18 **(b)** 9 **(c)** 6 **(d)** 36

9. The area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 6)$ is: 1

(a) 5 **(b)** 10 **(c)** 12 **(d)** 24

10. If $S_n = n^2 + 2n$, then $a_1 =$ 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

11. A man goes 15 m due east and then 20 m due north. His distance from the start (m) is: 1

(a) 625 **(b)** 5 **(c)** 35 **(d)** 25

12. If the midpoint of $(2, 3)$ and (x, y) is $(5, 6)$, then $x =$ 1

(a) 8 **(b)** 3 **(c)** $\frac{7}{2}$ **(d)** 12

13. The sum of first 6 positive odd numbers is: 1

(a) 42 **(b)** 30 **(c)** 36 **(d)** 72

14. In $\triangle ABC$, $AB = 9$, $BC = 12$, $AC = 15$. The triangle is: 1

(a) isosceles right **(b)** right-angled **(c)** not right-angled **(d)** equilateral

15. The distance of $(5, 3)$ from the y-axis is: 1

(a) 8 **(b)** 5 **(c)** 3 **(d)** 15

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Which term of the AP $3, 8, 13, 18, \ldots$ is 63? 2

17. A ladder 17 m long reaches a window 15 m high. Find how far the foot of the ladder is from the wall. 2

18. Find the distance between the points $(2, 3)$ and $(4, 1)$. 2

19. Find the sum of the first 10 multiples of 3. 2

20. Sides of a triangle are 7 cm, 24 cm, 25 cm. State with reason whether it is right-angled. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In an AP, the 2nd term is 7 and the 5th term is 19. Find the AP and its 10th term. 3

22. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 4$ cm, $DB = 6$ cm and $AE = 12$ cm, find EC. 3

23. Find the value of k for which the points $(1, 2)$, $(3, 6)$ and $(5, k)$ are collinear. 3

24. How many terms are there in the AP $18, \frac{31}{2}, 13, \ldots, -\frac{19}{2}$? 3

OR

The areas of two similar triangles are 16 cm² and 36 cm². If a side of the first is 4 cm, find the corresponding side of the second. 3

25. Find the area of the triangle with vertices $(1, 1)$, $(4, 5)$ and $(7, 2)$. 3

OR

Find the sum of the first 22 terms of the AP $15, 19, 23, 27, \ldots$ 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. How many terms of the AP $24, 22, 20, 18, \ldots$ must be taken so that the sum is 144? 5

OR

Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. 5

27. Find the point on the x-axis which is equidistant from $(7, 6)$ and $(-3, 4)$. 5

OR

A loan is repaid in 20 monthly instalments; the first is ₹100 and each later instalment is ₹10 more than the previous. Find the instalment in the last month and the total amount repaid. 5

END OF QUESTION PAPER — Total: 50 Marks